

Topic 12: Object-Oriented Programming — Classes and Objects

11.1 What is OOP?

- OOP organizes code around objects that combine data and the functions that work on that data
- Class: a blueprint or template — defines what data and actions an object will have
- Object: an instance of a class — the actual thing created from the blueprint
- A Car class defines: brand, speed, color (data) and accelerate(), brake() (functions)
- Your specific car is an object of the Car class

11.2 Defining a Class

```
class Student {  
public:  
    string name;  
    int marks;  
  
    void display() {  
        cout << name << ": " << marks << endl;  
    }  
};
```

- `class` keyword followed by the class name
- Members inside can be variables (attributes) or functions (methods)
- `public` means accessible from outside the class

11.3 Creating Objects

```
Student s1;  
s1.name = "Ahmad";  
s1.marks = 85;  
s1.display();  
  
Student s2;  
s2.name = "Sara";
```

```
s2.marks = 92;  
s2.display();
```

- Each object has its own copy of the attributes
- Dot operator `.` accesses members

11.4 Constructors

```
class Student {  
public:  
    string name;  
    int marks;  
  
    Student(string n, int m) {  
        name = n;  
        marks = m;  
    }  
};  
Student s1("Ahmad", 85);
```

- Constructor: a special function with the same name as the class, no return type
- Automatically called when an object is created
- Used to initialize attributes with values

11.5 Encapsulation — private vs public

```
class BankAccount {  
private:  
    double balance;  
public:  
    void deposit(double amount) {  
        if (amount > 0) balance += amount;  
    }  
    double getBalance() { return balance; }  
};
```

- `private`: accessible only inside the class — protects data from accidental change
- `public`: accessible from outside
- Encapsulation = hiding internal data and exposing controlled access through methods

11.6 Procedural vs OOP

- Procedural: `double getGrade(string name, int marks)` — data and functions separate
 - OOP: `Student s; s.getGrade();` — data and functions bundled together
 - OOP is easier to scale, maintain, and reason about in large programs
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